William Waddoups

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EDUCATION

Brigham Young University – Ira A. Fulton College of Engineering **Provo, UT**

Bachelor of Science in Mechanical Engineering *Sept. 2023 – Apr. 2028*

- Cumulative GPA: 3.96/4.00 – awarded BYU Academic Scholarship
- Engineering Honor Society Member – Tau Beta Pi

TECHNICAL SKILLS

- **Engineering Software:** SolidWorks, ANSYS, AutoCAD, Fusion 360, Inventor
- **Programming:** Python, C, NumPy, SciPy, Pandas, Matplotlib, Plotly
- **Hands-On:** Welding, Fabrication, Woodworking

WORK EXPERIENCE

Y-Emissions Laboratory **Provo, UT**

Research Assistant *May 2026 – Present*

- Designed and prototyped an emissions collection device using SolidWorks for moving semi-trucks with elevated exhaust pipes, applying engineering analysis while meeting project budget constraints
- Built a Python record-linkage pipeline matching 18,000+ on-road vehicle observations to VIN and EPA certification records, assigning emission control technology to 89% of the fleet
- Led quality assurance of the matched dataset via custom visualizations and summary statistics, identifying systematic errors that improved matching accuracy by 16%
- Co-authored "Heavy-Duty Diesel Vehicle Deterioration: Impact of Vehicle Age and Emission Control Technology on NOx Emissions," under review at *Environmental Science & Technology*

BYU Mechanical Engineering Department **Provo, UT**

Teaching Assistant for Leadership in Global Context *Apr. 2026 – June 2026*

- Taught full class sessions on leadership principles and coordinated with a team of TAs to support student development

LooLoo **Draper, UT**

Product Development Intern *Sept. 2025*

- Commissioned by CEO to research and develop a self-cleaning toilet seat concept for a future product line
- Presented functioning prototype to the executive team to evaluate product feasibility

Janicki Industries **Layton, UT**

Maintenance and Facilities Intern *May 2025 – Aug. 2025*

- Led the rebuild of 15 product transportation carts, identifying and implementing two upgrades to improve safety and durability while maintaining full compliance with Quality Assurance protocols
- Worked with the Maintenance Team to conduct weekly preventative maintenance on 6 mills and their 20+ support systems
- Assisted with critical component replacement on industrial 5-axis mills, including a C-axis gear box, high-pressure nitrogen line, drive magnets, pallet air bearings, and pallet station locators

Outlier AI **Remote**

Part-time Writer *Apr. 2024 – Aug. 2024*

- Evaluated AI-generated responses for factual accuracy, general effectiveness, grammatical quality, and user preference
- Wrote constructive feedback used by LLM developers to evaluate model performance and implement improvements

VOLUNTEER EXPERIENCE

The Church of Jesus Christ of Latter-day Saints **Budapest, Hungary**

Volunteer Representative *July 2021 – July 2023*

- Coordinated the immigration process in fluent Hungarian for 60+ volunteers, including compiling and organizing hundreds of legal documents, preparing detailed applications, and constant monitoring of permit status
- Dedicated 70+ hours a week to advancing missionary initiatives, serving the community, and training new volunteers

INTERESTS : Skiing deep powder, fly fishing for rainbow trout, playing new golf courses, and practicing traditional archery